FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 01

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Topics to be covered :

POLYNOMIAL & QUADRATIC EQUATION



Let $a \neq 0$

eg:		Degree	Domain
	$P(x) = kx^0$ (Constant Poly)	0	$(-\infty, \infty)$
	$P(x) = ax+b$ (Linear Poly)	1	"
	$P(x) = ax^2+bx+c$ (Quadratic Poly)	2	"
	$P(x) = ax^3+bx^2+cx+d$ (Cubic Poly)	3	"

$\downarrow$

$P(x) = 0.x^3 + ax^2+bx+c$ ✗ it is still a Poly

Domain → Set of Permissible Values of x for which func^n is defined known as Domain

for eg $f(x) = 3x+2$; $D_f = (-\infty, \infty)$, But for $f(x) = \frac{1}{x}$, $D_f = R - \{0\}$

Linear Poly

it is not a poly.

eg check which of the following are not polynomials

(1) $f(x) = 4x^2 - 2x + 1$ ✓ , (2) $f(x) = \frac{1}{3x-2}$, (3) $f(x) = x^{1/3} + 4$, (7) $f(x) = x$ ✓

× (Rational fun^n) × (Irrational func^n)

(4) $f(x) = \frac{3x^2 - 2x + 1}{4x + 5}$ (5) $f(x) = e^x$ (6) $f(x) = \log x$

× Rational func^n × Transcedental fun^n × T. fun^c

(7) $f(x) = |x| = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases}$ still it is not a poly, it is Piecewise func^n

Def^n Poly → (i) Domain = R, (2) Degree = W (3) Def^n of func^n Should be same for each value of x.

⊛ Any poly may have any number of terms (But finite)

eg $P(x) = 2x^{150} + 5x^{149} - 3x^{27} + 4x^{23} - 2x^{14} + 11x^2 - 2x + 4$ ↪ Dom $= (-\infty, \infty)$, Deg $= 150$

eg $P(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots\ldots\ldots$ (it is not poly ∵ it has Infinite No of terms)

⊛ General form of poly of degree 'n' → $P(x) = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{n-1} x^{n-1} + a_n x^n$

where $a_n \neq 0$ & Dom $= (-\infty, \infty)$

⊛ $P(x)$ is called Poly while $P(x) = 0$ is called Poly eqn's

for eg $P(x) = ax^2 + bx + c$ is Quad Poly while $ax^2 + bx + c = 0$ is called Q. Eqn

again $P(x) = ax^3 + bx^2 + cx + d$ is cubic Poly while $ax^3 + bx^2 + cx + d = 0$ ∴ Cubic eqn's

MONOMIAL → If any poly has only one term then it is called Monomial

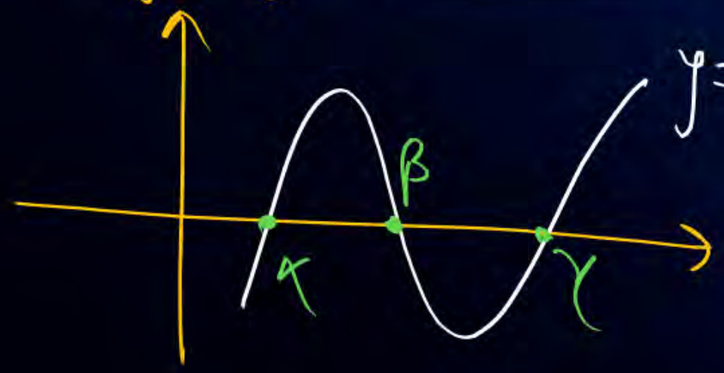
eg $P(x) = 4x^5$ (Monomial of degree 5), $P(x) = x^2$ (Monomial of degree 2)

BINOMIAL → If any poly has exactly 2 term then it is called BINOMIAL

eg $P(x) = 3x+2$ (Binomial of degree 1), $P(x) = x^5 - 3x^2$ (BINOMIAL of degree 5)

Roots of Poly eqn → (Roots / Soln / Zeros / Zero Crossings (Real Roots)) → Let us consider a

Poly eqn $\boxed{P(x) = 0}$ or $f(x) = 0$



$y = f(x)$

$y(\alpha) = f(\alpha) = 0$

$y(\beta) = f(\beta) = 0$

$y(\gamma) = f(\gamma) = 0$

i.e $f(\alpha) = f(\beta) = f(\gamma) = 0$

OR

$P(\alpha) = P(\beta) = P(\gamma) = 0$

Here $x = \alpha, \beta, \gamma$ are the Roots of $P(x) = 0$

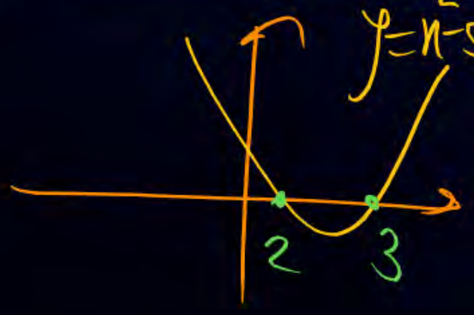
i.e those points where graph of $f(x)$ cuts X axis known as Roots

if [$x=\alpha$ is the Root of $P(x)$] $\Longleftrightarrow$ [$P(\alpha)=0$] OR then

Real Roots $\simeq$ Zero Crossings But Comp. Roots $\neq$ Zero Crossings

eg $P(x)=x^2-5x+6$ $\because$ $P(2)=0$, $P(3)=0$ $\therefore$ $x=2$ & 3 are Roots of $P(x)=0$

eg $P(x)=x^2+4$ $\because$ $P(2i)=4i^2+4=-4+4=0$, $P(-2i)=4i^2+4=-4+4=0$ $\therefore$ $x=2i$ & $-2i$ are the Roots of $P(x)=0$





⊛ Complex roots occurs in Conjugate pair if Coefficients are Real

for eg; $x^2-(3+5i)x+15i=0$

$x^2-3x-5ix+15i=0$

$x(x-3)-5i(x-3)=0$

$(x-3)(x-5i)=0$

$x=3,\ 5i$

Fundamental Th →

"n^{th} degree poly has exactly n Roots" (Real/Comp)

OR

A Poly (having degree ≥ 1) must have at least one root

eg $x^2-6x+9=0 \Rightarrow (x-3)^2=0 \Rightarrow x=3,3$

⊛ Constant Poly $P(x)=K$ does not have any root.

or $y=f(x)=k$



⊛ odd degree poly with <u>with Real Coeff</u> must have at least <u>one</u> real root. bcoz in that case Complex Roots (if exist) will occur in Pair

DESCARTE'S Rule → Consider the poly $P(x)$ then

(i) Number of +ve real roots $\leq$ Number of times sign changes in $P(x)$

(ii) " " −ve " " $\leq$ " " " " " $P(-x)$

eg $x^3-5x^2+7x-3=0$

Sol: Consider $P(x)=x^3-5x^2+7x-3$

No. of +ve Real roots $\leq$ three time

Now $P(-x)=-x^3-5x^2-7x-3$

No. of −ve real roots $\leq$ Zero time

while exact roots are $x=3,1,1$

eg $P(x)=x^3-6x^2+11x-6$

No. of +ve Real roots $\leq$ three

Now $P(-x)=-x^3-6x^2-11x-6$

No. of −ve Real roots $\leq$ Zero

while roots are $x=1,2,3$

eg $x^2-2x-15=0 \Rightarrow$ No. of +ve Real Roots $\leq$ one

$P(-x)=+x^2+2x-15=0 \Rightarrow$ No. of −ve " " $\leq$ one

& the roots are $x=-3,5$

eg: $P(x) = +x^5 - 6x^3 - 6x^2 - 7x - 6 \Rightarrow$ No. of +ve Real roots $\leq$ one

$P(-x) = -x^5 + 6x^3 - 6x^2 + 7x - 6 \Rightarrow$ No. of -ve Real roots $\leq$ four

$i^4 = 1,\ i^3 = -i,\ i^2 = -1$

while roots are $x = -1, -2, 3, +i, -i$

Cross check for i → $P(i) = i^5 - 6i^3 - 6i^2 - 7i - 6$

$= i(1) - 6(-i) - 6(-1) - 7i - 6 = 0$

eg. $P(x) \equiv x^9 - 5x^3 - x^2 + 7x + 2$

$P(-x) = -x^9 + 5x^3 - x^2 - 7x + 2$

Various possible Ans:-

this eqn has at Most two +ve roots

" " three -ve roots

& at least 4 Complex Roots

Cubic Equⁿ → it's General form is $ax^3+bx^2+cx+d=0$ (where $a\neq 0$)

Let α, β, γ are the roots of this equⁿ's (ie $P(\alpha)=P(\beta)=P(\gamma)=0$) then

$$\alpha+\beta+\gamma=\frac{-b}{a}, \quad \alpha\beta+\beta\gamma+\gamma\alpha=\frac{c}{a}, \quad \alpha\beta\gamma=\frac{-d}{a}$$

⊛ if α, β, γ are known then Cubic equⁿ can be written as,

$$x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-\alpha\beta\gamma=0$$

for eg: $x^3-5x^2+7x-3=0$

$(ax^3+bx^2+cx+d=0)$

$a=1, b=-5, c=7, d=-3$

& Roots are $\alpha=3, \beta=1, \gamma=1$

$\alpha+\beta+\gamma=-\left(\frac{b}{a}\right)=-\left(\frac{-5}{1}\right)=5$ ie verified

$\alpha\beta\gamma=\frac{-d}{a}=-\left(\frac{-3}{1}\right)=3$

2nd Part: if $\alpha=3, \beta=1, \gamma=1$ are the roots of any cubic eqn" then find it.

Sol: $x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-(\alpha\beta\gamma)=0$

$x^3-(5)x^2+(3+1+3)x-(3)=0 \Rightarrow x^3-5x^2+7x-3=0$

Q: if $\alpha=1, \beta=2, \gamma=3$ then find an eqn" for which above are the roots.

Sol: $\boxed{x^3-(\alpha+\beta+\gamma)x^2+(\alpha\beta+\beta\gamma+\gamma\alpha)x-(\alpha\beta\gamma)=0}$

$x^3-6x^2+(2+6+3)x-(6)=0 \Rightarrow x^3-6x^2+11x-6=0$

Method of Solving Cubic eqn → Solve $x^3-12x^2+39x-28=0$

factors of 28 $\longrightarrow$ 1, 2, 4, 7, 14, 28

So it's roots will be from $\pm1, \pm2, \pm4, \pm7, \pm14, \pm28$

Let $P(x)=x^3-12x^2+39x-28 \Longrightarrow P(1)=1-12+39-28=0$ ☺

ie $x=1$ is the root of given eqn's $\Longrightarrow$ $(x-1)$ will be the factor of $P(x)$

$P(x) \equiv x^3-12x^2+39x-28=0$

$x^3-x^2-11x^2+11x+28x-28=0$

$x^2(x-1)-11x(x-1)+28(x-1)=0$

$(x-1)\left[x^2-11x+28\right]=0$

$(x-1)\left[x^2-4x-7x+28\right]=0$

$(x-1)\left[x(x-4)-7(x-4)\right]=0$

$(x-1)(x-4)(x-7)=0$

$x=1, 4, 7$

(M-II) $P(x) \equiv x^3 - 12x^2 + 39x - 28 = 0$

$\because x = 4$ satisfies above eqn $\Longrightarrow (x-4)$ will be the factor of given eqn

$$x^3 - 12x^2 + 39x - 28 = 0$$

$$x^3 - 4x^2 - 8x^2 + 32x + 7x - 28 = 0$$

$$x^2(x-4) - 8x(x-4) + 7(x-4) = 0$$

$$(x-4)(x^2 - 8x + 7) = 0$$

$$(x-4)[x^2 - 7x - x + 7] = 0$$

$$(x-4)(x(x-7) - 1(x-7)) = 0$$

$$(x-4)(x-7)(x-1) = 0 \Rightarrow x = 1, 4, 7$$

Q: Solve $x^3-5x^2-18x+72=0$; $72 = 1, 2, 3, 4, 6, 8, 9, 12, 24, 36, 72$

Prof $\rightarrow$ ①

$\because P(1) \neq 0$, $P(2) \neq 0$, $P(3)=0$ ☺ $\Rightarrow x=3$ is the sol of ①

So By factor Th, $(x-3)$ is the factor of ①

$x^3-5x^2-18x+72=0$

$x^3-3x^2-2x^2+6x-24x+72=0$

$x^2(x-3)-2x(x-3)-24(x-3)=0$

$(x-3)(x^2-2x-24)=0$

$(x-3)[x^2-6x+4x-24]=0$

$(x-3)[x(x-6)+4(x-6)]=0$

$(x-3)(x-6)(x+4)=0$

$x=3, 6, -4$

Quadratic eqn :- Consider $P(x) = ax^2+bx+c$ or $\boxed{ax^2+bx+c=0}$ ——— (1)

Let α & β are the roots of this eqn then $\boxed{\alpha+\beta = -\frac{b}{a} \text{ & } \alpha\beta = \frac{c}{a}}$ same

Note: New eqn can be given as, $x^2-(\alpha+\beta)x+(\alpha\beta)=0$ ——— (2)

Verification :- $1.x^2-5x+6=0$ ——— (1)

$(ax^2+bx+c=0) \Rightarrow a=1, b=-5, c=6$

$x^2-2x-3x+6=0$

$x(x-2)-3(x-2)=0$

$(x-2)(x-3)=0$

$x=2,3$

Let $\alpha=2, \beta=3$

$\alpha+\beta = -\frac{b}{a} = -\left(\frac{-5}{1}\right) = 5$ is verified

$\alpha\beta = \frac{c}{a} = \frac{6}{1} = 6$ " "

Now using $x^2-(\alpha+\beta)x+\alpha\beta=0$

or $x^2-5x+6=0$ is verified

PQ:- Find the Q. equ^n for which Sum & Product of Roots are -3 & 2 resp.

eg ATQ, $\alpha+\beta=-3$ & $\alpha\beta=2$ then equ^n is $x^2-(\alpha+\beta)x+\alpha\beta=0$

or $x^2-(-3)x+2=0$ or $x^2+3x+2=0$

PQ: if α & β are the roots of $x^2-x+1=0$ then evaluate $\alpha^2+\beta^2=?$

Sol: $a=1, b=-1, c=1 \Rightarrow \alpha+\beta=-\frac{b}{a}=1$ & $\alpha\beta=\frac{c}{a}=1$

w.k.that $(\alpha+\beta)^2=\alpha^2+\beta^2+2\alpha\beta \Rightarrow \alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=1-2(1)=-1$

PQ: the roots of $(a-b)x^2+(b-c)x+(c-a)=0$ are ?

Sol: By observation, $x=1$ is the root of this equ^n (i.e. $\alpha=1$) $\quad Ax^2+Bx+C=0$

Now $\alpha\beta=\frac{C}{A}=\frac{c-a}{a-b} \Rightarrow 1\cdot\beta=\frac{c-a}{a-b} \Rightarrow \beta=\frac{c-a}{a-b}$ } Ans

Nature of Roots → Consider $ax^2+bx+c=0$ ↗ $\alpha+\beta=-b/a$, $\alpha\beta=c/a$

① if a & c has same sign & b has opposite sign ⟹ Both Roots are +ve

② if a, b, c have same sign ⟹ Both Roots are −ve

③ if a & c have opposite sign ⟹ one root is +ve and one root is −ve

eg $x^2-5x+6=0 \Rightarrow x=2,3$ (Both +ve) ∵ a & c have same sign & b has opposite sign

eg $x^2+5x+6=0 \Rightarrow x=-2,-3$ (Both −ve) ∵ a, b, c have same sign

eg $x^2+x-6=0 \Rightarrow x=2,-3$ (one +ve & one −ve) ∵ a & c have opposite sign

Method of solving Q.Eqn

$$ax^2+bx+c=0 \quad ; a\neq 0$$

$$a\left[x^2+\frac{b}{a}x+\frac{c}{a}\right]=0$$

$$\left[x^2+2\left(\frac{b}{2a}\right)x+\frac{c}{a}\right]=0$$

$$x^2+2\left(\frac{b}{2a}\right)x+\frac{b^2}{4a^2}-\frac{b^2}{4a^2}+\frac{c}{a}=0$$

$$\left(x+\frac{b}{2a}\right)^2-\left(\frac{b^2-4ac}{4a^2}\right)=0$$

$$\left(x+\frac{b}{2a}\right)^2=\frac{D}{4a^2}$$

$$x+\frac{b}{2a}=\pm\frac{\sqrt{D}}{2a}$$

$$x=-\frac{b}{2a}\pm\frac{\sqrt{D}}{2a}=\frac{-b\pm\sqrt{D}}{2a}$$

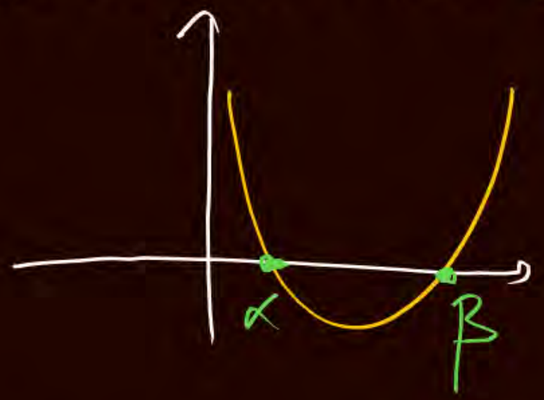
ie $\alpha=\frac{-b+\sqrt{D}}{2a}$ & $\beta=\frac{-b-\sqrt{D}}{2a}$

where $\boxed{D=b^2-4ac}$ = Discriminant

$y = ax^2 + bx + c$

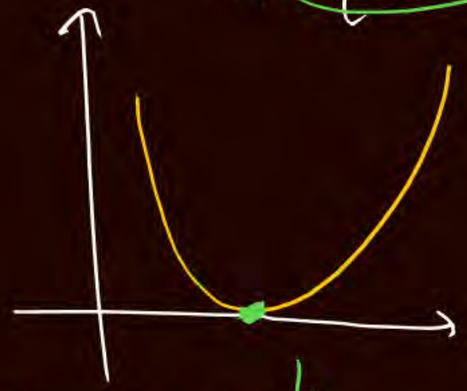
Note: ① if $a > 0$ & $D > 0$

⇒ Both the Roots are Real & distinct



② $a > 0$ & $D = 0$

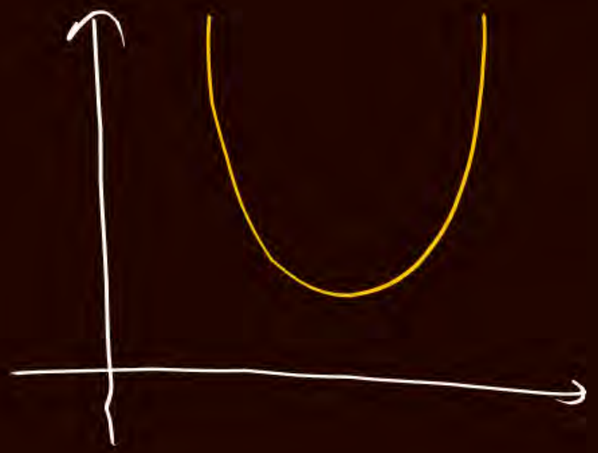
Both the roots are Real & equal



$\alpha = \beta = \frac{-b}{2a}$

③ $a > 0$ & $D < 0$

Both the roots are Imaginary



Q: The least +ve integral Value of k for which roots of the equ^n $x^2 + 5x + k = 0$ are Complex

Sol: $1 \cdot x^2 + 5x + k = 0$

$(ax^2 + bx + c = 0)$

$\Rightarrow a = 1, b = 5, c = k$

For Complex roots $D < 0$

$b^2 - 4ac < 0$

$25 - 4(1)k < 0$

$25 - 4k < 0$

$4k - 25 > 0$

$k > \frac{25}{4}$

$k > 6.25$

i.e $k \geq 7$

Ans $k = 7$

eg: $x^2+x-6=0$

$x^2+3x-2x-6=0$

$(x+3)(x-2)=0$

$x=-3, 2$

(M-II) $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

$=\frac{-1\pm\sqrt{(+1)^2-4(1)(-6)}}{2(1)}$

$=\frac{-1\pm\sqrt{25}}{2}=\frac{-1\pm5}{2}$

$x=2, -3$

Q: The Number of real roots of the eqn are ?

$(x-1)^2+(x-2)^2+(x-3)^2=0$

this eqn has NO real roots

bcoz eqn is not possible mathematically

(∵ Sum of three +ve Nos can't be zero)


Thank
YOU